



Revisiting group contribution theory for estimating fractional free volume of microporous polymer membranes

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ABSTRACT

Fractional free volume (FFV) is a commonly used metric for the development of structure–property relationships for polymer membranes. The most common method to calculate FFV uses Bondi's group contribution method, first introduced in 1964. While updated in 1997, there has not been a significant compilation of new structural motifs since the advent of linear microporous polymers. In this study, we critically examined the assumptions in Bondi's original method and provide four recommendations to streamline and improve the accuracy of calculating van der Waals volume (V_W) for any group. Using these recommendations, we created an updated list of V_W values for structural groups commonly present in microporous polymers. The V_W and FFV values were then calculated for a database of 123 microporous and high free volume polymers from the literature, showing an average 7% decrease in V_W and corresponding increase in FFV by a factor of 24% when compared to prior group contribution correlations in the literature. The significant apparent increase in estimated FFV provides a new perspective to understand and interpret the role of free volume on the separation performance of linear microporous polymers. Additionally, standardization of the group contribution method allows for the direct comparison of FFV values across studies.

1. Introduction

Fractional free volume (FFV) is widely used to describe the relative “empty space” present in a polymer membrane resulting from inefficient packing of the polymer chains in the solid-state [1]. Historically, this free volume concept has been used in the works of Doolittle [2–4] and Williams, Landel, and Ferry [5] to develop relationships between inherent viscosity and the glass transition temperature, respectively. Currently, FFV is used in a number of polymer applications, such as low- κ dielectrics [6], proton-exchange membranes [7], and gas separation membranes [8], as a fundamental material property to which changes in the polymer structure and performance could be related. For example, in the context of designing a polymer for gas separations, the incorporation of a bulky group on the polymer backbone could be hypothesized to increase FFV by disrupting chain packing and therefore be expected to result in higher gas permeability [9]. With such

observations, generalized correlations relating gas permeability to FFV can be made. Of note, a larger FFV has been shown to correlate with higher permeability coefficients in an exponential manner for a large database of reported polymeric materials [10]:

$$P = A \times \exp\left(-\frac{B}{FFV}\right) \quad (1)$$

where A and B are constants for a specific gas. In this way, FFV is a useful tool to understand and develop structure–property relationships to aid in the design of materials for specific applications.

However, determining FFV is empirically challenging. FFV is defined as

$$FFV = \frac{V_f}{V} = \frac{V - V_{oc}}{V} \quad (2)$$

where V_f is the free volume ($\text{cm}^3 \text{g}^{-1}$), V is the specific volume (cm^3

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g^{-1}), and V_{oc} is the volume occupied by the polymer chains ($\text{cm}^3 \text{g}^{-1}$) [10]. While the specific volume is a measurable quantity equal to the inverse of the polymer film density, researchers must turn to theoretical or empirical models in order to obtain an estimate of V_{oc} . The most widely used model for V_{oc} calculations in the gas separation membrane field was presented by Lee in 1980 [11], who proposed a method to obtain V_{oc} using a common correlation with the van der Waals volume (V_W):

$$V_{oc} = 1.3 \times V_W \quad (3)$$

where the 1.3 term is commonly interpreted as an inverse packing factor relating V_{oc} (i.e., the molar volume at 0 K) and V_W [12]. A comprehensive review on the origin and validity of this 1.3 factor is discussed at length by Horn [13] and is briefly discussed in this study as well.

On the other hand, V_W is estimated most commonly using Bondi's group contribution method [14]. In Bondi's seminal paper on this topic [14], the van der Waals (VDW) radii of atoms were compiled from X-ray diffraction measurements and were used to determine the atomistic volume contributions in molecules that are impenetrable to thermal collision, the so-called "van der Waals volume", which became the basis of group contribution theory. Bondi used a simple geometric method to determine the V_W of two bonded atoms. For example, V_W for a C–H group would be calculated by adding the V_W of the carbon and hydrogen atoms minus the intersection/overlapping volume of the spheres. The intersection volume can be exactly calculated and is a function of the radii of both spheres and the distance between their centers (i.e., the bond length) [15]. Therefore, with knowledge of the chemical structure, VDW radii, and bond lengths, one could feasibly construct any polymer by applying this geometric method for all chemical bonds in the polymer repeat unit. Bondi performed these calculations for common groups such as, $-\text{CH}_2-$, $-\text{CH}_3$, and $>\text{C}=\text{O}$, thereby creating a modular approach toward estimating V_W for more complex structures. Over time, the V_W of larger groups were calculated and tabulated, with those of Park and Paul and van Krevelen being the most cited by membrane researchers [12, 16].

However, the most recent compilations of Park and Paul and van Krevelen were published in 1997 and 2009, respectively [12,16]. Since the first report of PIM-1 as a gas separation membrane in 2005 [17], significant research has been expended in the synthesis of linear microporous polymers, resulting in hundreds of unique structures containing novel, contorted structural units (e.g., triptycene [18–20] or Tröger's base [21,22] motifs) or pendant groups (e.g., amidoxime [23] or methyl tetrazole [24]) [25–27]. As these new groups were not included in the previous compilations, manual calculation of V_W through atom-by-atom addition using Bondi's method becomes increasingly cumbersome.

The aim of this work is threefold, focusing on the calculation of V_W through group contribution theory and the resulting effect on FFV. First, the method for V_W calculation through Bondi's group contribution method was critically evaluated and an updated method with four recommended changes is presented. Second, using the updated method, V_W was calculated for both groups in previous compilations as well as for new groups and structural motifs commonly observed for modern linear microporous polymers. Lastly, the FFV values were calculated for a literature compilation of 123 microporous and high free volume polymers and various analyses were conducted to investigate the implications of the new FFV values on large-scale transport correlations for microporous polymers as a class of materials.

Recently, researchers have used molecular dynamics simulations to estimate FFV, wherein multiple polymer chains are relaxed within a specified box size and a probe is used to determine the amount of free volume [28]. While this approach allows for consideration of potential interchain interactions on the packing structure, these simulations can be computationally expensive especially for increasingly rigid structures and large repeat units [29]. Additionally, a previous study reported

similar FFV values estimated via MD simulations and calculated via Bondi's method, suggesting that Bondi's method can provide an acceptable estimate of FFV [28]. In this study, we follow Bondi's group contribution theory to estimate FFV, thus avoiding computationally-intensive simulations in lieu of a simpler but still accurate group contribution method with broad appeal.

2. Methods

2.1. Changes from Bondi's method

Examining Bondi's group contribution method, we identified the four following underlying assumptions with potential for refinement:

1. The VDW radii reported by Bondi represent values still accepted today.
2. Bond lengths can be fixed by bond type (e.g., a C–H bond is always 1.089 Å).
3. Overlap volume only needs to be considered for directly bonded atoms.
4. For unspecified atom-atom connections (i.e., connections between groups), V_W can be calculated by first assuming that the group is connected to an aliphatic carbon atom and then subtracting the sphere cap of the connecting atom.

The following sections describe our approach to address each assumption, with our primary changes noted in italics. Additionally, we assume hybridization does not have a significant effect on the VDW radius (i.e., sp^2 and sp^3 carbons have the same VDW radius).

Assumption 1. The VDW radii reported by Bondi represent values still accepted today.

Bondi's set of VDW radii is still the most widely used in the literature, but there exists slight differences in the VDW radii reported by Pauling in 1940 as well as Slonimskii et al. in 1970 [30,31]. Notably, in 1996, Rowland and Taylor examined a large database of crystallographic data reported since the 1970s to revisit the accuracy of these originally reported VDW radii [32]. They concluded Bondi's values showed a negligible discrepancy for all atoms except for hydrogen and suggested that 1.1 Å is more appropriate than the original value of 1.2 Å. *Following Rowland and Taylor's conclusions, we used a radius of 1.1 Å for hydrogen and Bondi's reported VDW radii for all other atoms.*

Assumption 2. Bond lengths can be fixed by bond type (e.g., a C–H bond is always 1.089 Å).

Bond lengths show some deviation depending on their local steric and electronic environments. For example, a C–C bond in a cyclopropane molecule is 1.526 Å, while a C–C bond in a cyclohexane molecule is 1.540 Å [33]. Since the intersection volume is a function of the bond length, these changes in bond length will affect the V_W calculation. *Thus, we implemented a simple molecular dynamics (MD) geometry optimization step to obtain atom coordinates for each group.* Bond lengths specific to the atom and its environment can then be determined directly from their coordinates.

Assumption 3. Overlap volume only needs to be considered for directly bonded atoms.

Bondi's method only considered subtraction of the intersection volume between directly bonded neighbors and does not account for the spatial orientation of the atoms. While this assumption accounts for the large majority of the intersection volume, the use of increasingly contorted structural groups present in state-of-the-art microporous polymers introduces the possibility of intersecting volumes between atoms that are not directly bonded due to their unusual, forced spatial orientation. Inclusion of the overlapping volumes between atoms that are not directly bonded will necessarily reduce the estimation of V_W , but likewise improve accuracy. For example, consider when atom 1 is bonded to

atom 2, and atom 2 is bonded to atom 3. Through Bondi's method, V_1 and V_2 would first be connected to form a volume V_a , then V_3 would be added but only considering the overlap between atoms 2 and 3:

$$|V_1 \cup V_2| = |V_1| + |V_2| - |V_1 \cap V_2| \equiv |V_a|$$

$$|V_a \cup V_3| = |V_a| + |V_3| - |V_a \cap V_3| = (|V_1| + |V_2| + |V_3|) - (|V_1 \cap V_2| + |V_2 \cap V_3|)$$

However, the union of three sphere volumes can be exactly described through the inclusion-exclusion principle as follows [15]:

$$|V_1 \cup V_2 \cup V_3| = (|V_1| + |V_2| + |V_3|) - (|V_1 \cap V_2| + |V_1 \cap V_3| + |V_2 \cap V_3|) + (|V_1 \cap V_2 \cap V_3|)$$

When taking the difference between the two methods, the Bondi method predicts V_W values that are greater than the exact volume by $|V_1 \cap V_3| - |V_1 \cap V_2 \cap V_3|$. Consequently, this extra volume is overcounted by the Bondi method and will result in an overestimation of the total V_W . To account for these additional terms, we estimated V_W through a Monte Carlo (MC) method, explained in detail below. If a large number of points is used, an accurate estimate of V_W can be obtained.

Assumption 4. For unspecified atom-atom connections (i.e., connections between groups), V_W can be calculated by first assuming that the group is connected to an aliphatic carbon atom and then subtracting the sphere cap of the connecting atom.

In Bondi's original work, the reporting of V_W without its corresponding sphere cap (see Supplementary Information for more details) was likely intended for ease of use such that one could build a complex structure by only needing to add the V_W of the groups required. However, as stated by Bondi, V_W was calculated assuming the group is connected to an aliphatic carbon atom. Since the intersection volume of spheres is a function of both the atom VDW radii and bond length, this approach is not necessarily appropriate depending on the structure. Thus, we have decided to report the V_W of groups with their specified sphere cap at the connecting atoms and provide a list of typical bond types with their corresponding overlapping volumes to subtract when adding groups together. While this approach makes estimation of V_W slightly more convoluted than simply adding groups, the accuracy is improved and the method is more modular. The effect of this assumption is visible particularly when considering heteroatom bonds, which will be discussed in depth later.

To summarize, we have implemented the following changes from Bondi's method with the aim of improving V_W calculations for any group of interest:

- The VDW radius of hydrogen was changed from 1.2 Å to 1.1 Å.
- MD simulations were used to perform geometry optimizations on structural groups to obtain atom coordinates.
- Given the atom coordinates and atom identity from the MD simulation, an MC simulation was used to estimate V_W .
- The V_W reported here does not assume a connection to an aliphatic carbon and does not exclude the sphere cap.

2.2. Procedure for simulated V_W of groups

The estimation of V_W through our proposed simulation method is a two-step MD simulation followed by an MC procedure. The group of interest was drawn in Materials Studio 8.0 with dummy methyl or aromatic groups representing a connection to another unspecified group. The dummy groups are used solely in this step for geometry optimization and are ignored in the MC portion of our procedure. The MD simulation

was performed using the Forcite module with the COMPASS II force-field, chosen for the extensive number of atom types and forcefield terms that are supported [34]. A Perl script was then used to extract the geometry-optimized atom coordinates and their corresponding identities. The coordinates and identity of the atoms were then transferred to a MATLAB script for the MC portion (see Supporting Information) and the atoms corresponding to the dummy groups were identified. In the MC script, a box was first created around the group and 1,000,000 points with random coordinates were generated within the confines of the box. For each point, the distance between the point and each atom was

calculated. If the calculated distance was less than the VDW radius of the atom, the point was considered to be "inside the group" and the script would move to the next point. By running a large number of randomly generated points, an unbiased calculation of V_W and its standard deviation (σ) can be estimated using the ratio of points "inside the group" to the total number of points [15]:

$$V_{group} = V_{box} \times \frac{n_{inside}}{n_{total}} \quad (4)$$

$$\sigma_{group} = \sqrt{V_{group} \times \frac{(V_{box} - V_{group})}{n_{total}}} \quad (5)$$

where V_{group} is the volume of the group ($\text{\AA}^3$), V_{box} is the volume of the simulation box ($\text{\AA}^3$), n_{inside} is the number of points considered to be "inside the group", and n_{total} is the total number of points tested, equal to 1,000,000 for the simulations presented here. V_{group} can then be converted to the more conventional $\text{cm}^3 \text{mol}^{-1}$ form using Avogadro's number. The V_W calculated for various groups are tabulated in Tables 1–3.

The overlapping volume of typical bond types (e.g., C–C, C–O, C_{ar}–O, etc.) were determined by drawing the simplest structural unit containing the bond type of interest in Materials Studio 8.0, then performing the MD simulation as described above. Using the calculated distance between the two atoms of interest, the intersecting volume was calculated with knowledge of the atom identities. A full list of bond types and their corresponding overlapping volumes are tabulated in Table 4.

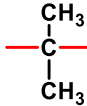
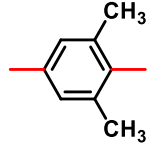
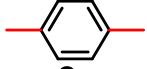
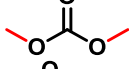
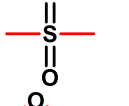
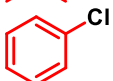
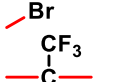
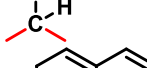
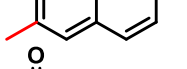
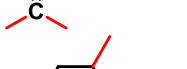
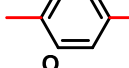
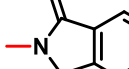
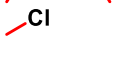
3. Results

3.1. Simulated V_W values for previously reported and new structural groups

Following the procedure outlined in Section 2.2, new V_W values were simulated for the 41 groups originally presented by Park and Paul [16] (Table 1) and 32 new groups that are commonly found in microporous and other high-free volume polymers (Table 2). Additionally, for convenience, 23 of the most common groups found in microporous polymers (e.g., triptycene) were simulated as well (Table 3). These large structural groups can be built from the smaller groups in Tables 1 and 2, but using the larger groups improves the accuracy when applying this method. Our newly reported groups were chosen such that all polymers featured in recent publications of microporous polymers could be built. In Tables 1–3, the groups of interest are represented by black atoms. However, we also show red atoms that are not included in the V_W values. These atoms represent the structure of neighboring groups, such as aliphatic or aromatic connections. Therefore, the red atoms represent dummy groups used solely for geometry optimization. Additionally, Table 4 contains information about the bond lengths and calculated overlap volume for each bond type, and Table 5 shows the atoms

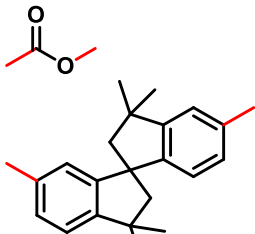
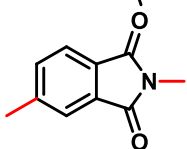
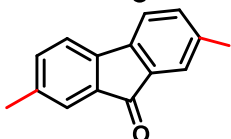
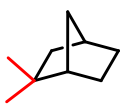
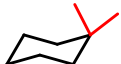
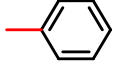
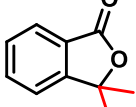
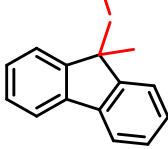
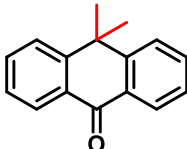
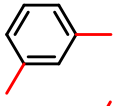
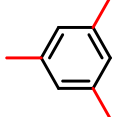
Table 1

Reference ID, structure, simulated V_W , and standard deviation in MC simulation for the groups previously considered by Park and Paul [16]. Red bonds directly attached to black atoms represent connections to neighboring groups with the specified geometry. In some cases, extended red structures (e.g., aromatic groups) are shown to more clearly indicate appropriate connecting groups. V_W includes molar volume of the sphere cap(s) at atoms that can connect to other groups.

Updates for groups previously reported by Park and Paul			
Reference ID	Structure	V_W (cm ³ mol ⁻¹)	σ (cm ³ mol ⁻¹)
P1		32.58	0.08
P2		64.5	0.1
P3		46.25	0.08
P4		25.25	0.05
P5		23.14	0.04
P6		8.51	0.02
P7		13.54	0.02
P8		15.97	0.02
P9		51.6	0.1
P10		24.39	0.05
P11		44.67	0.09
P12		13.92	0.03
P13		71.9	0.1
P14		16.24	0.03
P15		45.48	0.08
P16		92.7	0.2
P17		14.63	0.04
P18		19.37	0.03
P19		13.53	0.02
P20		20.63	0.04

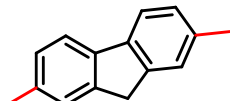
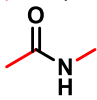
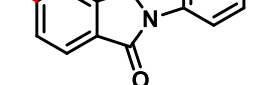
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Table 1 (continued)

Updates for groups previously reported by Park and Paul			
Reference ID	Structure	V_W (cm ³ mol ⁻¹)	σ (cm ³ mol ⁻¹)
P21		159.7	0.4
P22		69.5	0.1
P23		93.5	0.1
P24		58.7	0.1
P25		54.6	0.1
P26		14.73	0.02
P27		47.11	0.09
P28		9.42	0.02
P29		13.15	0.03
P30		42.5	0.1
P31		66.7	0.1
P32		91.5	0.2
P33		102.4	0.2
P34		12.39	0.02
P35		46.19	0.08
P36		45.44	0.08

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Table 1 (continued)

Updates for groups previously reported by Park and Paul			
Reference ID	Structure	V_W (cm ³ mol ⁻¹)	σ (cm ³ mol ⁻¹)
P37		91.1	0.2
P38		22.05	0.04
P39		11.20	0.03
P40		3.35	0.01
P41		107.9	0.2

considered and their respective radii and VDW volumes.

An immediate comparison can be made between the V_W values tabulated by Park and Paul against those presented in Table 1 (see groups P1 to P41) [16]. The V_W values simulated using the new method appear to be greater than those of Park and Paul by 25% on average, but this increase is mostly artificial and relates primarily to changes in methodology related to Assumption 4. The procedure by Park and Paul reports V_W values for groups without their sphere cap(s). Conversely, our reporting of V_W includes the sphere cap(s), resulting in the higher apparent values. Thus, when separating groups P1 to P41 by valency, the monovalent, divalent, trivalent, and tetravalent groups showed an apparent average V_W increase of 12%, 15%, 37%, and 128%, respectively. Viewed this way, there is a general trend of artificially larger V_W values corresponding to the number of sphere caps added per valency. Of course, group-to-group structural variation, the number of hydrogen atoms, and the presence of heteroatoms at valency points all influence V_W , but those specific factors are concealed by inclusion of sphere caps.

Given this methodology change, the new method requires subtraction of the overlap volume between the atoms that connect groups. When assembling groups into a polymer structure, the red atoms should be representative of the structure of the neighboring connecting group, but only the overlapping volume between the directly bonded atoms needs to be subtracted according to Equation 6:

$$V_{W,tot} = \sum_{i=1}^n V_{W,i} - \sum_{j=1}^m V_{overlap,j} \quad (6)$$

where n is the number of groups, m is the number of connections between groups, $V_{W,i}$ is the van der Waals volume for the groups from Tables 1–3, and $V_{overlap,j}$ is the overlap volume between groups from Table 4.

In this way, the V_W for a polymer repeat unit can be calculated. As one example, group P6 has methyl dummy groups, representing single-bonded connections to neighboring groups. The overlap volume between group P6 and its connecting groups is subtracted in the V_W calculation. In a case where P6 is connected to P3 on both sides (i.e., P3–P6–P3), the overlap volume corresponding to a C_{ar}–O bond (C8) would be subtracted twice. As another example, S2 shows a common spiro center, but the reported V_W can only be used when this spiro center is connected to two aromatic rings on neighboring groups. As in the previous example, the overlap volume between the groups will likewise be subtracted. In a case where S2 is connected to P11 on both sides for a total of four connections (i.e., P11>S2<P11), the overlap volume corresponding to a C–C_{ar} bond (C4) would be subtracted four times. Additionally, it should be noted that we do not strictly enforce implicit

hydrogen substitutions on the dummy aliphatic and aromatic groups. Any substitution on the neighboring groups that differs from the implicit hydrogens on the dummy group is assumed to have little effect on the overlap volume.

As an aside, when constructing polymers containing aromatic connections, some groups in Table 1 can be treated as roughly equivalent, such as groups P3 and P35 (bivalent, para versus meta connections, 46.249 cm³ mol⁻¹ versus 46.187 cm³ mol⁻¹) or P15 and P36 (trivalent, 1, 2, 4 versus 1, 3, 5 connections, 45.484 cm³ mol⁻¹ versus 45.438 cm³ mol⁻¹). In both cases the simulated V_W values are effectively the same, suggesting a general heuristic that only the number of connections matters for V_W calculations as opposed to the location of the connections for aromatic rings. For example, in the case of an aromatic ring containing ortho connections in the repeat unit, one could use a V_W value of 46.2 cm³ mol⁻¹ as a reasonable estimate for that group. In a similar manner, groups that are identical except for possessing different connection bond types (e.g., P7 and P19 showing aliphatic and aromatic chlorine groups) show effectively the same V_W because any changes in connection bond types are accounted through the use of overlap volume specific to the bond type.

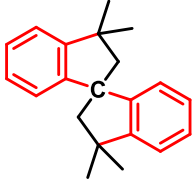
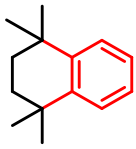
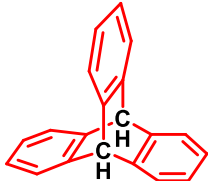
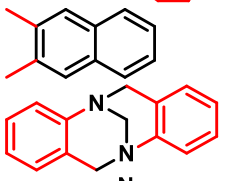
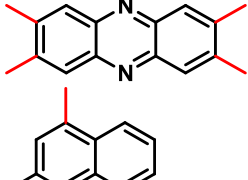
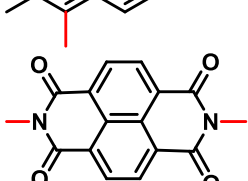
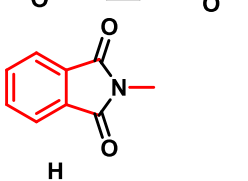
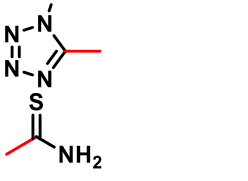
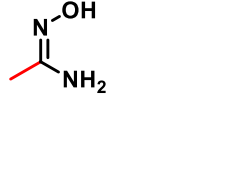
3.2. Updated FFV estimates for literature microporous polymers

New V_W values were calculated for a database of 123 polymer structures that had a reported density. The polymers included in the database are all solution-processable and are either microporous (i.e., have a Brunauer-Emmett-Teller (BET) surface area greater than 30 m² g⁻¹, a typical value for common glassy polymers) [35] or define various upper bounds (i.e., commercial perfluoropolymers) [36]. The full database is presented in Table S1. For 7 polymers in the database, the reported FFV was calculated using MD simulations instead of using group contribution methods. These polymers are denoted in Table S1 with a “*”. Since the FFV values obtained using group contribution and MD simulations are not directly comparable, these polymers were not included in any analysis when comparing reported FFV from group contribution methods.

The reported density was used with the new V_W values to calculate FFV. Example calculations from our group contribution method are included in the Supplementary Information. It should be noted that for the same polymer backbone (i.e., polymers possessing the same V_{oc}), variations in the recalculated FFV may occur due to differences in the reported density. For example, the new FFV for PIM-1 ranges from 0.248 to 0.414. Differences in the reported density can originate from variations in synthesis approach (e.g., low temperature [37] versus high temperature [38] PIM-1 synthesis), treatment prior to measurement (e.

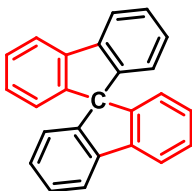
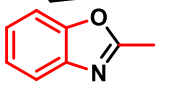
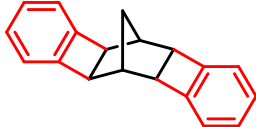
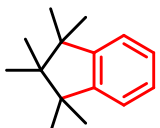
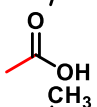
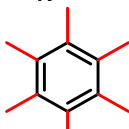
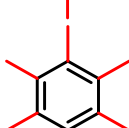
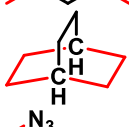
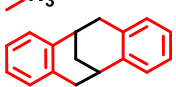
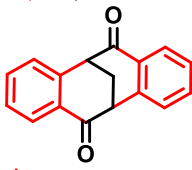
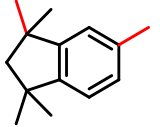
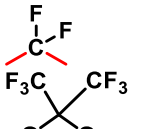
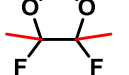
Table 2

Reference ID, structure, simulated V_W , and standard deviation in MC simulation for the new groups considered in this study. Red bonds directly attached to black atoms represent connections to neighboring groups with the specified geometry. In some cases, extended red structures (e.g., aromatic groups) are shown to more clearly indicate appropriate connecting groups. V_W includes molar volume of the sphere cap(s) at atoms that can connect to other groups.

Small groups (previously unreported)			
Reference ID	Structure	V_W (cm ³ mol ⁻¹)	σ (cm ³ mol ⁻¹)
S1		16.57	0.03
S2		86.8	0.2
S3		78.8	0.2
S4		25.32	0.06
S5		72.2	0.1
S6		44.2	0.1
S7		89.6	0.2
S8		80.8	0.2
S9		118.3	0.2
S10		33.08	0.06
S11		31.52	0.06
S12		29.22	0.06
S13		29.88	0.07
S14		95.6	0.2

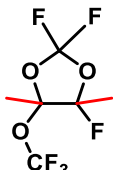
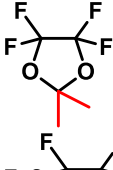
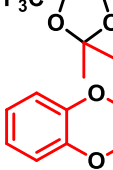
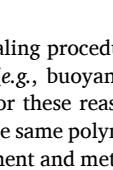
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Table 2 (continued)

Small groups (previously unreported)			
Reference ID	Structure	V_W (cm ³ mol ⁻¹)	σ (cm ³ mol ⁻¹)
S15		21.86	0.04
S16		58.0	0.1
S17		87.6	0.2
S18		21.60	0.04
S19		40.70	0.09
S20		43.10	0.07
S21		43.91	0.08
S22		39.91	0.09
S23		19.64	0.04
S24		48.7	0.1
S25		53.8	0.1
S26		96.0	0.2
S27		20.62	0.04
S28		82.3	0.2

(continued on next page)

Table 2 (continued)

Small groups (previously unreported)			
Reference ID	Structure	V_W (cm ³ mol ⁻¹)	σ (cm ³ mol ⁻¹)
S29		70.9	0.1
S30		50.7	0.1
S31		66.4	0.1
S32		17.00	0.04

g., methanol treatment, different annealing procedures, etc.) as well as the method used to measure density (e.g., buoyancy in a liquid [39], geometrically calculated [40], etc.). For these reasons, care should be used when comparing FFV values for the same polymer backbone across studies since these differences in treatment and methodology will affect the density measurement.

A plot comparing the reported FFV to the new FFV is shown in Fig. 1. It should be noted that not all polymers in the database have a reported FFV and would therefore not be represented in Fig. 1. In almost every case, with the notable exception of the commercial perfluoropolymers, the new FFV was greater than the reported literature FFV by a factor of 24%, on average. This increase in FFV indicates a decrease in V_W for each polymer resulting from the method changes described in Section 2.1. Using the reported FFV values as a basis, changing from the original to the new method resulted in roughly a 7% decrease in V_W .

In evaluating the effect of each modified assumption on V_W , it was found that the decrease in the hydrogen VDW radius (Assumption 1) was a moderate contributor to lower V_W , while variations in bond lengths and consideration of non-adjacent overlap volumes (Assumptions 2 and 3, respectively) had a relatively minor effect. Using the spirobisindane (SBI) group as an example (see group L3 from Table 3), a V_W of 157.7 cm³ mol⁻¹ was estimated through the new method. When Assumption 1 is relaxed (i.e., V_W calculated using a hydrogen VDW radius of 1.2 Å), the V_W shows a 5% increase up to 166.3 cm³ mol⁻¹. To evaluate the impact of Assumption 2, the bond lengths present in the SBI unit were determined from MD simulations and the average length for each bond type (i.e., C_{ar}-C_{ar}, C_{ar}-C, C_{ar}-H, C-C, and C-H) was determined. When comparing to literature compilations [41], the deviation in average bond length ranged from -0.3% for C_{ar}-C bonds to 1.5% for C_{ar}-H bonds, corresponding to 0.4% and -2.2% changes in the overlap volume, respectively. The impact of Assumption 3 can be evaluated by counting the number of points that were located exclusively in the intersection of two non-bonded atoms. In the SBI example, roughly 0.4% of the points satisfied this condition, indicating minimal over-counted overlap volume if the original Bondi method were used. Of note, our use of MD simulations results in Assumptions 2 and 3 being coupled because various repulsion terms present in the forcefield minimize overlap of non-bonded VDW volumes. Taken together, Assumptions 1 to 3 had a small to moderate impact on V_W values.

On the other hand, changes related to Assumption 4 affected V_W values for polymers containing a large number of heteroatom bonds,

particularly in the case of the perfluoropolymers. A detailed explanation is provided in the Supplementary Information. Briefly, the assumption that all groups are connected to an aliphatic carbon atom results in a size mismatch at the sphere-sphere cross-sectional interface for heteroatom bonds because of differences in the bond length. For perfluoropolymers, this assumption resulted in a larger V_W , which in turn led to smaller FFV. In contrast with all other polymers considered, the high-free volume commercial perfluoropolymers (Teflon AF 2400, Teflon AF 1600, and Hyflon AD 60) showed a 14% increase for calculated V_W and a corresponding 17% decrease in FFV.

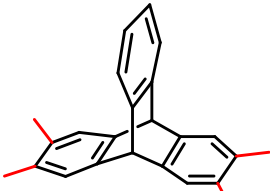
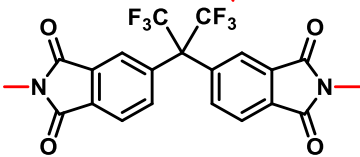
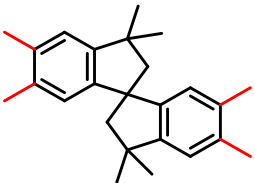
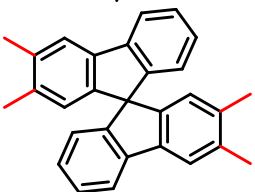
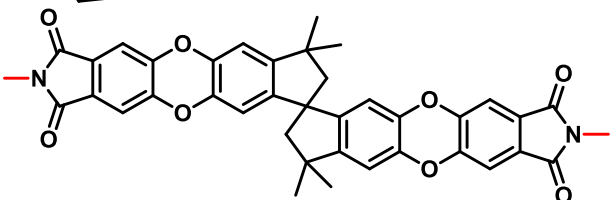
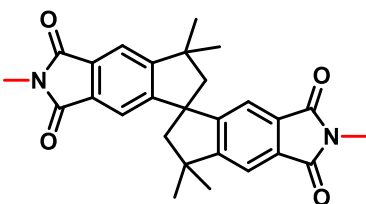
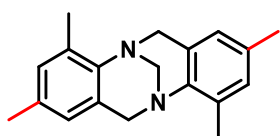
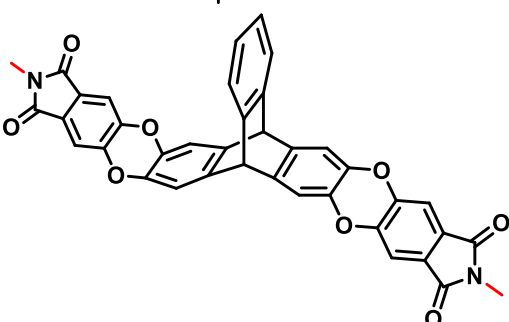
3.3. Impact of new FFV values on gas transport correlations for microporous polymers

3.3.1. Permeability correlations with respect to FFV

Correlations between FFV and general classes of microporous polymers were evaluated using Equation (1). These correlations are exclusively based on our newly developed method and 120 microporous polymers in our database. The perfluoropolymers were not included for this analysis since they do not show measurable BET surface areas and are therefore not considered to be microporous [42]. An example of this analysis is shown in Fig. 2a for CO₂. Similar plots are shown for all other gases in Fig. S6. Black symbols correspond to the reported literature FFV and red symbols correspond to FFV calculated using our newly developed method. There exist a number of black symbols that deviate significantly from the correlation in Equation (1). However, when FFV is calculated using the new method, these original outlier points experience a relatively larger increase in FFV, resulting in an overall improved correlation; for CO₂, the Pearson correlation coefficient changes from -0.48 to -0.67 when comparing the old versus new FFV correlations. Similar changes were observed for all gases. All slopes and Pearson correlation coefficients are tabulated in Table S2. Interestingly, a large number of these original outlier points are PIM-1 or PIM-1-like polymers. This finding indicates two possibilities. First, the original method may contain limitations in accuracy for estimating FFV. Additionally, the large variations in FFV for identical polymers may result from user error in needing to piece together non-intuitive groups to assemble polymer repeat units. It is likely that both of these possibilities contribute to our findings, emphasizing the great need to modernize and simplify group contribution methods for microporous polymers. Use of our group contribution method for all polymers in the database provides

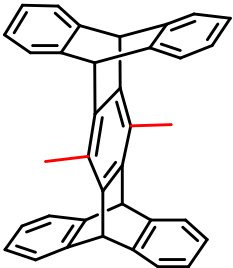
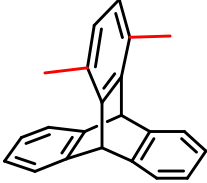
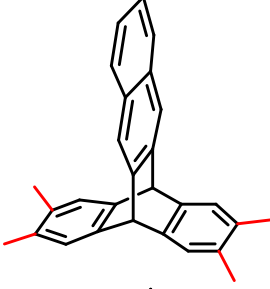
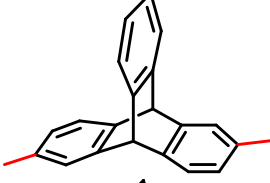
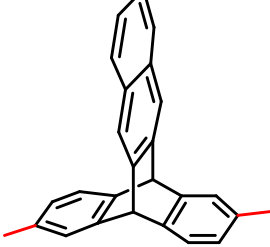
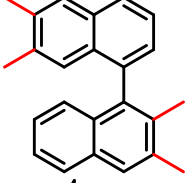
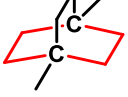
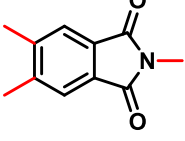
Table 3

Reference ID, structure, simulated V_W , and standard deviation in MC simulation for the large groups considered in this study. Red bonds directly attached to black atoms represent connections to neighboring groups with the specified geometry. In some cases, extended red structures (e.g., aromatic groups) are shown to more clearly indicate appropriate connecting groups. V_W includes molar volume of the sphere cap(s) at atoms that can connect to other groups.

Large groups (previously unreported)			
Reference	Structure	V_W (cm ³ mol ⁻¹)	σ (cm ³ mol ⁻¹)
L1		133.5	0.3
L2		180.6	0.4
L3		157.7	0.4
L4		167.2	0.4
L5		297	1
L6		206.1	0.5
L7		135.8	0.3
L8		275.2	0.9

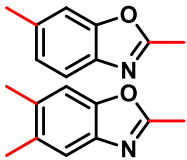
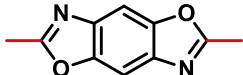
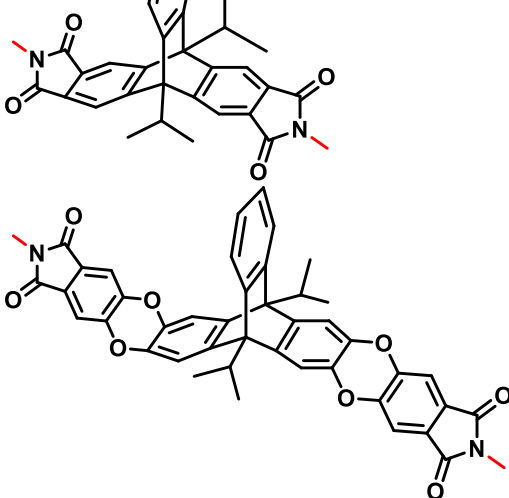
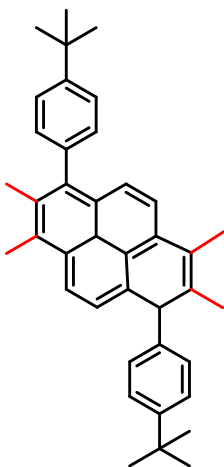
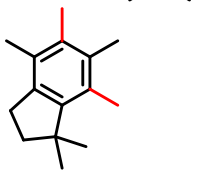
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Table 3 (continued)

Large groups (previously unreported)			
Reference	Structure	V_W (cm ³ mol ⁻¹)	σ (cm ³ mol ⁻¹)
L9		224.0	0.6
L10		135.2	0.3
L11		159.8	0.4
L12		135.5	0.3
L13		161.4	0.4
L14		137.2	0.3
L15		58.5	0.2
L16		68.7	0.1
L17		59.4	0.1

(continued on next page)

Table 3 (continued)

Large groups (previously unreported)			
Reference	Structure	V_W (cm ³ mol ⁻¹)	σ (cm ³ mol ⁻¹)
L18		58.5	0.1
L19		72.6	0.1
L20		238.1	0.6
L21		331	1
L22		262.0	0.8
L23		105.8	0.3

a consistent route for V_W calculation, and thereby acts as a normalization method to facilitate direct comparisons of FFV across multiple studies and research groups.

It should be noted that a moderate amount of scatter about the best-fit line is still observed for these correlations, even after normalizing the calculation method for V_{oc} for all polymers. This scatter can be explained in two ways. The first relates to uncertainties in reported densities, as discussed earlier, which will contribute to scatter in these plots. The

second, and likely more important, relates to factors affecting permeability that are not captured by FFV, such as differences in sorption interactions or free volume distribution. For example, a polymer that shows high FFV and low permeability could possess large pockets of free volume but have bottlenecks with limited free volume connectivity, thereby limiting diffusion. Alternatively, a polymer with low FFV could show particularly high sorption affinity to CO₂, thereby resulting in higher permeability. Considered in this way, the presence of scatter

Table 4

Reference ID, bond type, bond length, and calculated overlap volume for connections considered in this study.

Reference ID	Bond type	Bond length (Å)	Overlap volume (cm ³ mol ⁻¹)
C1	C-H	1.099	2.607
C2	C _{ar} -H	1.099	2.608
C3	C-C	1.525	4.614
C4	C-C _{ar}	1.508	4.688
C5	C _{ar} -C _{ar}	1.437	5.002
C6	C=C	1.337	5.462
C7	C-O	1.416	3.877
C8	C _{ar} -O	1.361	4.091
C9	C=O	1.209	4.694
C10	O-H	0.952	2.418
C11	C-N	1.455	4.018
C12	C _{ar} -N	1.395	4.255
C13	N-H	1.004	2.434
C14	C-S	1.807	3.973
C15	C _{ar} -S	1.777	4.100
C16	S-H	1.321	2.361
C17	C-F	1.380	3.840
C18	C _{ar} -F	1.341	3.988
C19	C-Cl	1.783	3.802
C20	C _{ar} -Cl	1.712	4.099
C21	C-Br	1.939	3.683
C22	C _{ar} -Br	1.920	3.763
C23	C-Si	1.885	5.336
C24	C _{ar} -Si	1.873	5.394
C25	Si-O	1.636	4.970

Table 5

Reference ID, van der Waals radius, and calculated van der Waals volume for atoms considered in this study.

Reference ID	Atom	VDW Radius (Å)	VDW Volume (cm ³ mol ⁻¹)
A1	H	1.1	3.357
A2	C	1.7	12.393
A3	O	1.5	8.513
A4	N	1.55	9.393
A5	S	1.8	14.711
A6	F	1.47	8.013
A7	Cl	1.75	13.519
A8	Br	1.85	15.971
A9	Si	2.1	23.361

demonstrates a weakness of FFV as a correlating variable for permeability from a transport perspective. However, given the significant reduction in scatter from our updated method, there is an advantage to applying this method for the development of improved structure-property-performance correlations.

A significant change was also observed for the best-fit correlation slope, B . Fig. 2b shows the obtained B values plotted against a measure of penetrant volume (diffusion diameter cubed [43]) for microporous polymers using both reported literature FFV as well as the FFV calculated from the new method. A linear relationship is observed for both cases, although there is a slight difference in the slope. According to the free volume theory of Cohen and Turnbull and assuming a negligible role of FFV on the sorption coefficient, B corresponds to a minimum critical volume required for diffusion with a γ scaling factor representing an average count of the number of molecules in a free volume element [43,44]. However, this theory was developed for rubbery polymers, so it is interesting that the linear relationship was preserved for these microporous materials. While further speculation is outside the scope of this work, future studies of this linear trend as well as interpretations of how γ changes for rubbery, glassy, and emerging microporous polymers would be of interest to elucidate the applicability of transport models across various classes of polymeric materials as free volume increases.

3.3.2. Revisiting packing factor assumptions

Two useful relationships related to V_w calculations were revisited in

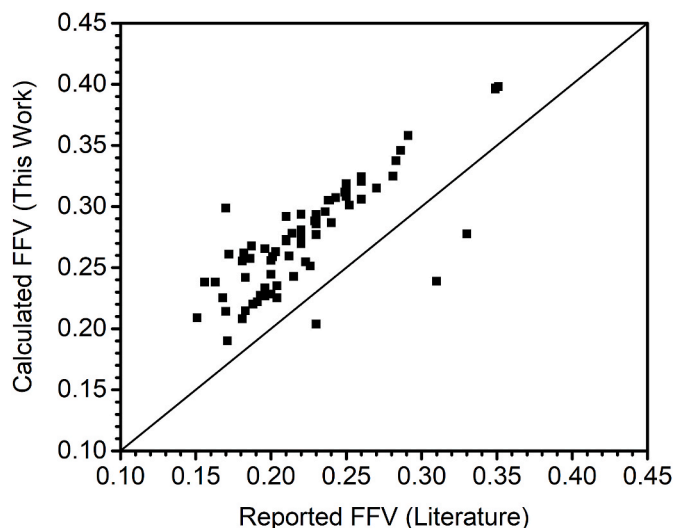


Fig. 1. Comparison of reported literature FFV values with calculated FFV values using the new method.

the context of microporous polymers. First, we will consider the relationship between specific density and V_w , as discussed by van Krevelen and Park and Paul [12,16]:

$$V = \beta \sum_{k=1}^K (V_w)_k \quad (7)$$

Where $\beta = 1.60$ at 25 °C for conventional glassy polymers (e.g., poly-sulfone, Matrimid, etc.), effectively acting as a room temperature packing factor to correlate the polymer density to its V_w [12]. Because glassy polymers are in a non-equilibrium state, β can be interpreted as a rough measure of how close the polymer is to its relaxed equilibrium state, with a larger value indicating a state further from equilibrium. This 1.60 value was obtained using the original Bondi method to calculate V_w [12]. Accounting for the 7% decrease in V_w between the Bondi method and the new method, a β value of 1.72 ± 0.04 can be estimated for conventional glassy polymers. Using the recalculated V_w values for microporous polymers, the average β value was found to be 1.8 ± 0.1 at 25 °C. The marginally higher average value of β for microporous polymers as a class suggests a higher degree of non-equilibrium character compared to typical glassy polymers, consistent with the ultra-rigid nature of the chemical motifs in microporous materials. Moreover, this finding correlates with polymers that have extremely high or undetectable T_g values [45,46] and accelerated physical aging trends [47,48], indicating that other theoretical correlations may exist to connect this work more broadly to phenomena in the polymer physics community.

Second, we will consider the relationship between V_{oc} and V_w (see Equation (3)), specifically concerning the 1.3 value for the inverse packing factor. As previously mentioned, there has been discussion regarding the origin of this value [13]. In particular, Horn concluded that absent other information, that 1.3 is a suitable approximation for the packing factor [13]. Here, the overall permeability versus $1/\text{FFV}$ correlations developed in Section 3.3.1 can be used to back-calculate the packing factor for each gas-polymer pair.

$$P = A \times \exp \left(- \frac{B}{1 - \alpha \times \frac{\rho V_w}{MW}} \right) \quad (8)$$

Where A and B are the best-fit intercept and slope of the P vs. $1/\text{FFV}$ correlations, α is the packing factor, ρ is the polymer density (g cm⁻³), V_w is the VDW volume (cm³ mol⁻¹), and MW is the molecular weight of

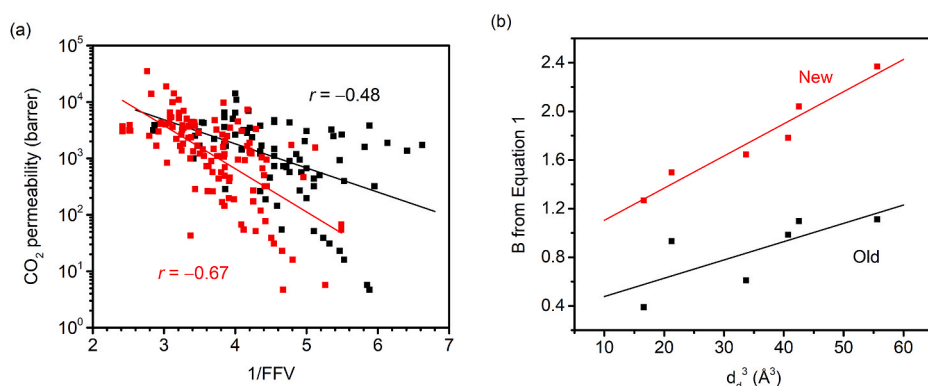


Fig. 2. (a) Example correlation of $\ln(P)$ vs $1/\text{FFV}$ for CO_2 with their respective best-fit lines. (b) Correlation between the best-fit B values from Equation (1) and diffusion diameter cubed. For both plots, the black symbols represent values obtained from reported FFV values of microporous polymers and the red symbols represent FFV values calculated using the new method. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the repeat unit (g mol^{-1}). Given the best-fit A and B values for each gas discussed in Section 3.3.1., an α value can be calculated for each gas–polymer pair. While this is an indirect method of obtaining the packing factor from large-scale transport correlations, this approach provides a valuable means to verify or refine this coefficient, analogous to how upper bound correlations have been used to refine gas diameter sizes [43]. Averaging the back-calculated α values for each gas–polymer pair in the database, a value of 1.3 ± 0.1 is obtained for α , in agreement with the typically used value. However, a standard deviation of 0.1 represents a significant variation in FFV. For example, the FFV of a PIM-1 polymer with a density of 1.1 g cm^{-3} would range from 0.245 to 0.353 within the range of $\alpha = 1.4$ to $\alpha = 1.2$. Interestingly, the primary contributor to this uncertainty for all gas–polymer pairs appears only when comparing α values across polymers. When considering the average α value obtained for a single polymer averaged across its gases, a small standard deviation on the order of ± 0.02 is observed, indicating a relatively consistent α value for each polymer. For example, the refined α value for PIM-1 from Chen et al. [49] is 1.28 ± 0.01 . While a value of 1.3 ± 0.1 for α looks to be appropriate for microporous polymers as a whole, there appears to be evidence supporting the existence of an individual α value for each polymer derived from its density and transport properties.

4. Conclusions

Estimation of FFV via the group contribution method was reexamined and applied to more recently developed microporous polymers. To calculate V_W , the assumptions and deficiencies present in the original method proposed by Bondi were examined. The following four changes were implemented to improve the accuracy of the V_W calculation and streamline calculations for new groups present in the current membrane literature: (1) The VDW radius of hydrogen was changed from 1.2 Å to 1.1 Å, (2) MD simulations were used to perform a geometry optimization on the structural group to obtain atom coordinates, (3) MC simulations were used to estimate V_W based on atom coordinates and atom identity from the MD simulations, and (4) V_W was reported with a sphere cap at the end group and without the assumption that the end group is connected to an aliphatic carbon. Using the new method, V_W values were obtained for a total of 96 groups commonly found in microporous and other high-free volume polymers. FFV was recalculated for a database of 123 microporous and high free volume polymers in the membrane literature, resulting in an average 7% decrease in V_W and a corresponding increase in FFV by a factor of 24% when compared to the originally reported literature values. The change in the hydrogen VDW radius accounted for a moderate fraction of the change in V_W , the MD and MC simulations accounted for a small fraction of the change, and assumptions related to the sphere cap accounted for the remainder. Correlations relating gas permeability to $1/\text{FFV}$ were updated using the new FFV values, resulting in a significantly improved correlation compared to the original FFV approach. The updated and modernized

method introduced here provides a streamlined process for straightforward and accurate V_W calculations, through which FFV can be calculated using group contribution theory. Additionally, standardization of the V_W calculation established through our method allows for the direct comparison of FFV values across studies. The recalculated FFV values and correlations obtained for microporous polymers provide a more accurate interpretation of the role of free volume on transport properties.

CRedit authorship contribution statement

Albert X. Wu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft. **Sharon Lin:** Conceptualization, Methodology, Validation, Investigation, Data curation, Writing – review & editing. **Katherine Mizrahi Rodriguez:** Conceptualization, Methodology, Validation, Investigation, Data curation, Writing – review & editing. **Francesco M. Benedetti:** Investigation, Data curation. **Taigyu Joo:** Investigation, Data curation. **Aristotle F. Grosz:** Investigation, Data curation. **Kayla R. Storme:** Investigation, Data curation. **Naksha Roy:** Investigation, Data curation. **Duha Syar:** Investigation, Data curation. **Zachary P. Smith:** Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2021.119526>.

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